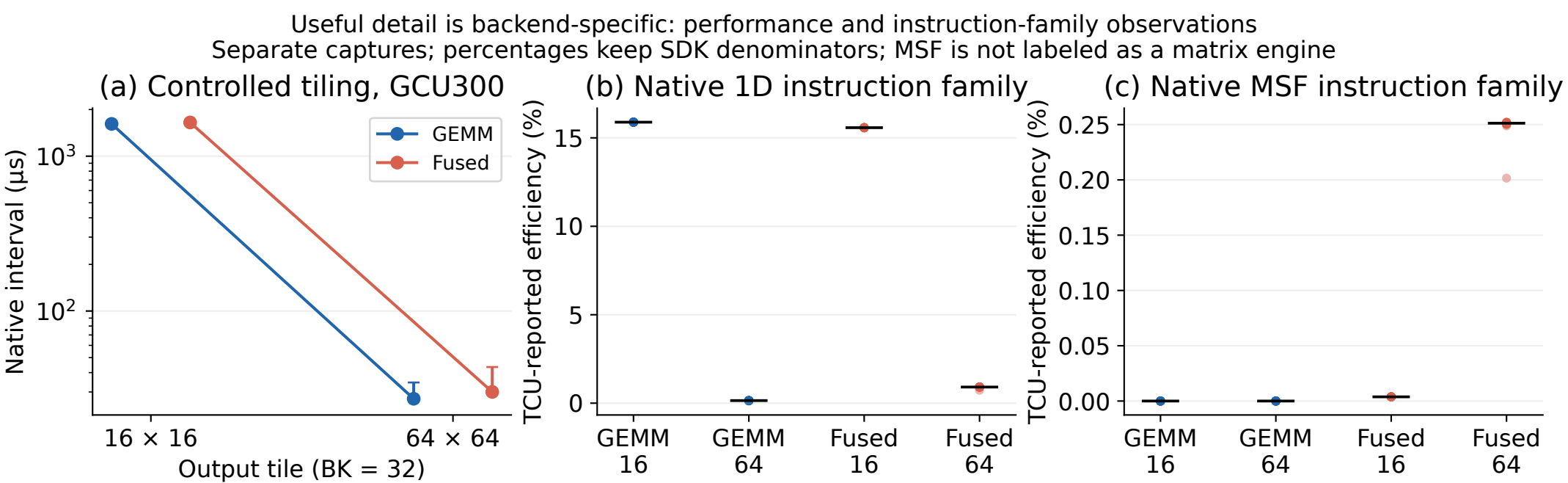
Useful detail is backend-specific: performance and instruction-family observations
Separate captures; percentages keep SDK denominators; MSF is not labeled as a matrix engine
(a) Controlled tiling, GCU300
Native interval (µs)
10^3
10^2
GEMM
Fused
16 × 16
64 × 64
Output tile (BK = 32)
(b) Native 1D instruction family
TCU-reported efficiency (%)
15
10
5
0
GEMM 16
GEMM 64
Fused 16
Fused 64
(c) Native MSF instruction family
TCU-reported efficiency (%)
0.25
0.20
0.15
0.10
0.05
0.00
GEMM 16
GEMM 64
Fused 16
Fused 64